Research memo: the arithmetic-geometric direction

PRIMES-Conjectures project — AG track, branch `arith-geom-interpretation` — October 21, 2026
Full development: `doc/arith-geom.tex` (v1.0, 15 pp); results log: `doc/arith-geom-findings.md`

TL;DR. Reading Agrawal's conjecture through algebraic groups turned the project's empirical dead ends (A11, N1, A10) into theorems, produced the first quantified evidence on Popovych's conjecture beyond search ranges, closed the elliptic-curve escape route, and gave the negative-result paper its structural spine: a single *stabilizer-index theorem* explaining why every variant of the attack bottoms out at the same twin-smooth wall.

1. The dictionary

The AKS congruence $(X-1)^n \equiv X^n - 1 \pmod{n, X^r-1}$ is a *Frobenius-lift test*: $\mathbb{Z}[X]/(X^r-1)$ is the Λ-ring of μ_r with Adams operations $\psi^k(X) = X^k$, and n is prime iff ψ^n is a Frobenius lift at n (Wilkerson); AKS checks the lift condition on the single section $X-1$. The exception clause $n^2 \equiv 1 \pmod r$ says exactly that the fake Frobenius σ_n lies in $\mathrm{Gal}(\mathbb{Q}(\zeta_r)/\mathbb{Q}(\zeta_r)^+) = \{1, \mathrm{conj}\}$ — the *stabilizer* of the section's divisor. This "clause = stabilizer" principle recurs in every result below.

On the counterexample side, the Lenstra–Pomerance conditions $p-1 \mid n-1$, $p+1 \mid n+1$ say that n *impersonates the Frobenius of* p *on both one-dimensional algebraic tori over* $\mathbb{F}_p$ (split: order $p-1$; nonsplit norm-one: order $p+1$). Twin-smoothness of $(\frac{p-1}{2}, \frac{p+1}{2})$ is torus-order smoothness — the same condition behind the B-SIDH/SQIsign parameter hunts, which is why the project's reliance on the Costello–Naehrig corpora was never a coincidence.

2. The four results

(AG2) The torus dimension-count theorem — A11 is now a corollary. For $d = \mathrm{ord}_r(p)$, the unique involution of $(\mathbb{Z}/r)^\times$ being -1 forces the exact *inversion collapse* $(X-1)^{p^{d/2}-1} = -X^{-1}$ (even d), so the control order $\rho_r(p)$ lives on cyclotomic levels $e \mid d'$, $d' = d/2$ or d, and the smoothness burden is the torus dimension $\sum_{e \mid d'} \varphi(e) = d'$. Validity ($\mathrm{ord}_r(n) > 2$) forces $d \geq 3$, so **burden ≥ 2 for every AKS modulus r, with the minimum 2 attained exactly at $d = 4$, levels $\{1,2\}$: the double (twin-smooth) condition** — realizable iff $4 \mid r-1$, and Váňa's $\rho_5(p) \mid 10(p^2-1)$ is the special case. The burden formula reproduces the A11 empirical table at all 11 moduli (value and minimizing d); verified on 67 sample primes. "Pick a friendlier r" is closed structurally, not just empirically.

(AG5) Popovych is strictly harder — first quantified evidence. The collapse above is a property of $X-1$, whose zero $\{1\}$ is Galois-stable. The second Popovych section $X+2$ has *trivial* stabilizer — $(\zeta^\gamma+2)/(\zeta+2)$ is not even a unit of $\mathbb{Z}[\zeta_r]$ for $\gamma \neq 1$ (disjoint ideal supports: norms r vs $(2^r+1)/3$) — so no collapse exists at any depth and the pair pays full burden $\min\{d : d \mid r-1, d \geq 3\} \geq 3$, equal to 4 at Popovych's own $r = 5$: pool primes would need $p-1$, $p+1$ *and* p^2+1 simultaneously smooth. At the frozen harvest spec that extra ≈102-bit level is a Dickman tax of 10^{-7}–10^{-11}, emptying the measured pool. Bonus structure: the second congruence's $X{=}1$ component is exactly the base-3 Fermat test $3^n \equiv 3$, and mod 3 the two sections coincide.

(AG3) The impersonation hierarchy — our targets have a name, and the wall has a shape. Define cells by (level set, Frobenius powers); at level 2 the power's parity is a sign. Then: Carmichael = $(\{1\}, 0)$, infinite (AGP); Lucas–Carmichael = $(\{2\}, 1)$, infinite (Wright 2018); the L–P targets = $(\{1,2\}, (0,1))$ are the *Williams numbers* (Williams 1977, Echi) — open even for nonemptiness, none below 2^{64} (McIntosh–Mitra 2015). Yet Howe's *rigid order-2 Carmichael number* 443,372,888,629,441 = 17·31·41·43·89·97·167·331 occupies the *same level set* with all-even parities and exists at 4.4×10^{14} (re-verified). **Sign splitting, not depth, is the wall**: rigid cells are one-sided AGP problems (single target 1); the Williams parities $(0,1)$ force targets $(+1,-1)$ that conflict on shared prime powers, forcing the coprime side-split whose measured collapse is N1/A10. Validity pins Agrawal to the sign-split cell; sharp datum: Howe's $n \equiv 1 \pmod 5$ does not even pass $T(-1, n, 5)$ — rigidity misses the μ_{2r} torsion the mod-80 conditions exist to fix.

(AG4) The elliptic axis is closed — the stabilizer-index theorem. A canonical elliptic analogue of $X \mapsto X^n$ exists only for CM curves (Λ-structure iff CM, Borger–Gurney). There Frobenius acts on an r-eigenline by a scalar λ, curve automorphisms give exact collapses (e.g. $x([i]S) = -x(S)$ on $y^2 = x^3 - x$), and everything

unifies:

$$\textbf{burden} \;=\; [\langle\lambda\rangle : \langle\lambda\rangle \cap U] \;\geq\; 2 \quad \text{for valid } \lambda \notin U,$$

U the unit-stabilizer of the section ($\{\pm 1\}$ for $X-1 \to$ AG2; $\{1\}$ for $X+2 \to$ AG5; μ_4/μ_6 for CM x-coordinates). The tantalizing elliptic burden-*1* cells ($\lambda \equiv \pm i$, section order collapsing to $p-1$ alone) are exactly the exception clause — the elliptic mirror of rigid Carmichaels — and valid burden-2 cells demand twin-smoothness again, with sign-split targets. Measured on $y^2 = x^3 - x$: valid-cell floors 3 (r=13) and 2 (r=17), matching the index formula; elliptic $\geq$ torus burden at all 11 moduli. Gate AG-G2 **not passed**; the negative result extends to the CM-elliptic axis.

3. Epistemic status

Proved outright: the collapse identities, the support bounds, the independence/stabilizer lemmas, the index formula, the burden minimizations. One ingredient is heuristic (Artin-type *generic sharpness*: the support bounds are attained), empirically supported with 0 exceptions across 196 sampled large primes — and it is a *theorem* in the function-field mirror via Weil's RH for curves (AG6 appendix), locating the residual difficulty over $\mathbb{Z}$ in the unconditional-equidistribution gap. All results are mechanism-scoped (Frobenius-matching constructions — every published counterexample approach); four deterministic verification scripts (`code/ag{2,3,4,5}_*.py`) with committed JSON back every claim.

4. Where this leaves the program

- **Negative-result paper**: written (branch `negative-result-writeup`, 11 pp) with the AG results as §5 — the obstruction is now structural, not parametric: no modulus, no second section, no CM curve escapes the stabilizer-index floor, and the floor's minimum is exactly the twin-smooth sign-split cell that N1/A10 measured dead.
- **Standalone note candidate**: AG5 (Popovych) is self-contained and would be the first published evidence in either direction beyond search ranges.
- **The only live constructive direction** remains option (b): a fundamentally new solver for 0/1 subset-products in $\sim$24k-bit smooth-exponent groups at identity-density $\sim$0.004. The AG results sharpen its stakes: a breakthrough would unblock the torus route only — Popovych and elliptic variants stay strictly harder.
- **Open problems surfaced**: unconditional sharpness (equidistribution); existence of Williams numbers at all (independent interest; the sign-split lens suggests why the AGP machine hasn't reached them); whether any side-separating polynomial family reaches an admissible local cell (old Q6.2, still open).